

Conformal Prediction is Early Stopping

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Abstract

Split conformal prediction ends where its calibration begins: fit a model, pool its held-out residuals, quote the pooled quantiles, stop. Of 686 arXiv papers with conformal prediction in the title, 599 stop at that empirical map, and the one candidate for continuing past it, on reading, also stops. We show the stop is the expensive part. An identity pins the irreducible excess log score of a frozen scheme to its score and stratification; on 15,022 economic series, stopping costs 0.15 to 0.18 nats net of estimation, composing past the map recovers 40 to 75 percent, and coverage detects none of it.

1 Introduction

Conformal prediction attaches a guarantee to any point forecaster by a recipe. Fit a model however you like. On held-out data, compute a scalar error for each point, for instance the absolute residual. Take the 90th percentile of those errors and quote the model’s prediction plus or minus that percentile. If the held-out data and the future point are exchangeable, meaning their joint law is unchanged by shuffling, the quoted interval contains the truth 90% of the time, at any sample size, with no assumption on the model or the data distribution.

The recipe ends with the percentile. Nothing is ever fitted to the output of the pooled-error step: the interval it produces is the answer. In machine learning, early stopping is a deliberate act, a regularizer chosen because validation loss says further fitting would hurt. The conformal stop is not chosen against anything. It is where the recipe happens to end, and as far as we can determine no paper argues for it.

This paper treats the stop as an empirical question. The unit throughout is the nat, the natural unit of the logarithmic score: a forecaster whose average log density trails another’s by 0.15 nats assigns the realized outcomes $e^{0.15} \approx 1.16$ times less probability per observation. Beside it we report CRPS, the continuous ranked probability score, which integrates the squared difference between the forecast’s cumulative distribution and the outcome’s step function; it is in the units of the data, lower is better, and it is the score most conformal papers report.

The paper has three components.

1. *The count* (Section 2). In a frame of 686 arXiv papers, 599 stop at the empirical map. Papers that attack the pooled quantile replace it or reweight it; none keeps it and continues. The single flagged candidate for continuing, a PNAS paper, conditions before the map and stops after it.

2. *The floor* (Section 3). For a class of frozen conformal schemes, the excess log score over the conditional oracle decomposes exactly into a conditional mutual information fixed by the score and stratification, plus an estimation term. Calibration choices move only the estimation term. The identity says what stopping protects and what it abandons.
3. *The price* (Section 4). Inside a transform grammar where the empirical map is one invertible step among peers, we hold one forecaster and one empirical object fixed and move only the object’s position: last, as the convention has it, or interior, with modelling continuing on its output. On 15,022 economic series, stopping costs 0.15 to 0.18 nats net of estimation; the interior placement recovers 40 to 75 percent of that depending on how much history the pooled window discards; and empirical coverage of the stopped arm is the best in the table, so the certificate is blind to the entire loss.

Section 5 states the recommendation: keep the map, keep the certificate where a certificate is wanted, and keep modelling past it.

What is new and what is borrowed. The count and the price are this paper’s, and the floor identity is proved in Appendix A with its scope stated carefully. The published theory around the pooled shape is context, not a contribution here: exact distribution-free conditional coverage is impossible in nonatomic settings (Lei and Wasserman, 2014; Barber et al., 2021); weighting calibration points unequally costs coverage in proportion to the total-variation distance between their residual laws (Barber et al., 2023); and a guarantee held at a fixed point rather than on average inflates the threshold for classical testing-power reasons (Li, 1989; Low, 1997; Cai and Low, 2004). Together these give the pooled shape a defence. Nothing in them concerns the stop, which is the point of this paper. The base forecaster is the subject of Cotton (2026).

2 The count

The claim that everyone stops concerns published work, so it gets a protocol: a defined frame, documented keyword rules, and full-text reading of every paper whose label could change the conclusion. The frame is every arXiv paper with “conformal prediction” in the title at harvest date, $n = 686$. Classifier, frame, and per-paper verdicts are in the repository.

The object classified is the *empirical map*: the step turning the multiset of held-out errors into a distribution, concretely the empirical cumulative distribution of the calibration scores. A paper *stops* if the output of that map, its quantiles, its p -values, or its intervals, is the answer. The alternatives found in the frame: *replace* the pooled quantile with a conditional estimate (SPCI fits a quantile regression to recent residuals (Xu and Xie, 2023)); *tune* the level online while keeping the map (adaptive conformal inference (Gibbs and Candès, 2021)); *consume* the output in a downstream decision such as screening pseudo-labels; or *compose*: keep the map and fit a model to its output, the move whose absence is the finding.

Of 686 papers, 599 stop, 16 replace, 28 tune, 38 consume, and 5 propose no method. One compose candidate was flagged by the screen: distributional conformal prediction (Chernozhukov et al., 2021), which uses an estimated conditional distribution as its score. Read in full, it conditions *before* the map, and the answer is the empirical quantile of the resulting ranks: a richer score, then a stop. The refutation criterion for the count is stated and falsifiable: a paper that retains the empirical rank transform, applies it to residuals, and fits a downstream model on

the transformed stream. Reweighting the empirical law does not count, conditioning upstream does not count, adapting the level does not count.

Outside the conformal banner the composing move is routine. The normal quantile transform in hydrological forecasting (Bogner et al., 2012), copula time-series models (Chen and Fan, 2006), and autoregressive transformation models (Rügamer et al., 2023) all apply an empirical marginal map and keep modelling in the transformed coordinates. The absence is a habit of one community, not a difficulty of the idea.

A second classification, reported in the repository, checks how papers treat residual pooling: 94.5% never let the weights on calibration residuals depend on the test point at any granularity, and each of the nine smooth-weighting papers that claims exact validity was read in full and found to claim it through a legitimate escape (marginal recalibration, a stated coverage gap, a known covariate-shift tilt, or added smoothness assumptions). The field polices the pooling constraint carefully. The stop it does not police at all.

3 The floor

The identity below says what the calibration step can and cannot influence. It needs two definitions and has a deliberately narrowed scope.

A *nonconformity score* is a map $A : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$ assigning a scalar to each candidate outcome y at input x . A *stratification* $\kappa : \mathcal{X} \rightarrow \mathcal{K}$ assigns each input to a category. Both are fit on a training fold and frozen, since exchangeability of calibration and test scores is what the coverage argument uses. The schemes covered by the proposition are those whose score is, for each x , a strictly increasing absolutely continuous bijection in y : signed-residual and normalized signed-residual schemes, and predictive systems built on invertible score transformations. Folded scores such as the absolute residual $|y - \hat{\mu}(x)|$ and quantile-regression scores are not bijections in y and need a separate treatment; the empirical arms in Section 4 are of the covered type.

Such a scheme emits the pooled empirical law of calibration scores in the test point’s stratum, pushed back through $A(x, \cdot)^{-1}$. For log scoring the emitted law must have a density, so let $\hat{G}_\kappa$ be an absolutely continuous estimate of the pooled score law (a kernel-smoothed or interpolated empirical law; the raw empirical measure is atomic and scores $-\infty$). Write $W = A(X, Y)$, let G_κ be the population pooled law, and let the benchmark be the conditional oracle, whose expected log score is the conditional entropy $H(Y | X)$.

Proposition 1 (The floor, exactly). *Fix the training fold, so (A, κ) is deterministic, and assume the laws below admit densities with finite divergences. Then the excess log score of the conformal predictor decomposes exactly as*

$$L(\hat{F}) - H(Y | X) = I(W; X | \kappa) + \mathbb{E}_\kappa \text{KL}(G_\kappa \| \hat{G}_\kappa),$$

where $I(W; X | \kappa)$ is the conditional mutual information between score and input given the stratum. The second term is nonnegative and is the only part any calibration choice, estimator, quantile convention, randomization, or level, can affect; it vanishes only under KL-consistent estimation of the pooled law with growing calibration mass per stratum, which is an assumption on the estimator, not a consequence of the setup.

The proof is a change of variables in which the Jacobian of A cancels between truth and forecast, then a split at the population pooled law (Appendix A). In words: the input reaches the predictive only through the frozen reshaping and the frozen stratum choice, so whatever

information about the score the input still carries within a stratum is lost before calibration begins, and no amount of calibration data buys it back. With a location score and one stratum the floor is $I(R; X)$, the mutual information between the model’s error and its input.

Remark 2. The floor is zero exactly when the frozen pair renders the score independent of the input within strata, that is, when the modelling was finished before the conformal step. Conformal calibration is deployed precisely when one is unwilling to assume that. The method’s stated reason for existing is its lower bound.

The floor explains what stopping protects: the certificate, which concerns the marginal rank of the test score and is genuinely exact at its position. It also says what stopping abandons: everything in $I(W; X | \kappa)$, plus whatever a downstream model could extract from the map’s output stream. The published results cited in Section 1 defend the pooling constraint inside the certificate. No result defends ending the pipeline. Whether ending it is expensive is a measurement question, taken up next.

4 The price

4.1 A laboratory where position is a variable

Inside standard conformal practice the stop is invisible: the pipeline ends at the map by construction, so there is nothing to vary. To measure the stop one needs a setting where the conformal step is an ordinary component, and that requires seeing it as what Section 3 showed it to be: a univariate transform. The raw empirical map is a step function, not invertible and without a density; repaired by strictly monotone interpolation through its knots, with kernel-smoothed mass so it can be log-scored, it becomes an invertible state-updated map of the observation coordinate, and such a map can sit anywhere in a chain.

The laboratory is a transform grammar (Cotton, 2026). A forecaster is a *chain* of state-conditional invertible transforms, each removing one kind of structure and passing its residual on: a location tracker subtracts a running mean, a scale transform divides by a running deviation, a difference removes trend. A *terminal leaf* models the law of whatever residual remains, and the predictive in the original coordinate is the leaf’s law pushed back through the inverse chain, change of variables handled exactly. The base forecaster, `laplace`, runs roughly sixty candidate chains in parallel, combines their predictive densities with likelihood weights updated every observation, splices generalized Pareto tails onto the extremes, and scores its own probability integral transforms online. It ranks first in the public benchmark this study extends, and its own terminal leaf is tuned to CRPS.

In this laboratory the conformal habit is one configuration among two:

stop. Take the finished predictive, keep its centre, replace its shape with the pooled empirical law of the model’s own recent errors, kernel-smoothed. Quote that. Nothing acts downstream. This arm is a frozen conformal scheme of the covered type, so Proposition 1 applies with floor $I(R; X)$.

compose-after. Install the identical empirical object as the terminal leaf *inside* the chain: the candidate pool still weights it against alternatives, the tail splice still repairs its extremes, and the transform chain still inverts it to the observation coordinate. This arm carries no coverage certificate. It is the move the floor identity permits, a change of representation, executed after the map rather than before it.

Both positions run flat and recency-weighted (exponential weights, half-life 150 observations), so staleness explains no comparison.

4.2 Design

The corpus is FRED economic change series (differences of rates, production, employment and similar), non-price, with two tiers. The *breadth* tier uses 256 observations of history per forecast origin and every qualifying series: 15,022 at this draft (daily and monthly arms complete, weekly still filling). The *deep* tier uses 1,024 observations and series with at least 1,100, so the 400-residual window is a strict subset of the history it could have used; its current panel is 212 series and is being enlarged. Twelve rolling origins per series, horizons 1, 6 and 12. Because origins within a series overlap, all intervals are bootstrap over series, not rows.

A control corpus of synthetic independent draws (Gaussian, Student- t_3 , Laplace) runs under the identical protocol at matched context. On the control the residual law does not depend on the state, pooling is correct, and a constrained arm should lose nothing; what it loses there is the cost of estimating a 41-node empirical law rather than a parametric scale, and that estimation cost is subtracted from every headline number below.

4.3 The decomposition, net of estimation

Breadth tier, 15,022 series, control-subtracted, log score deficit against the unconstrained base:

	$h=1$	$h=6$	$h=12$
stop, total cost	0.147	0.166	0.181
of which pooled shape	0.084	0.111	0.105
of which position	0.063	0.055	0.076

Raw per-arm deficits carry series-bootstrap 95% intervals of about ± 0.02 throughout. On the deep tier’s current 212-series panel the same decomposition is 0.58 total with 0.44 from position: when the pooled window is a strict subset of a long history, position dominates the shape constraint roughly three to one, and when the window already holds every residual (breadth tier), position and shape contribute comparably. One sentence organizes both tiers, and the enlarged deep tier will test it: *the price of stopping grows with how much the pooled window discards*, history it cannot hold and downstream machinery it cannot reach.

Per-series win rates sit between 43% and 51%: on the median series the arms are close, and the mean cost is carried by the minority of series where stopping hurts badly. Stopping behaves like the removal of insurance.

Adaptivity does not substitute for position. Recency-weighting the stop arm’s window recovers 0.02 to 0.07 nats across tiers and horizons, against 0.06 to 0.44 for moving the map’s position; and level adaptation is dispatched by Proposition 1 directly, since it acts on the pooled scores alone.

On CRPS the tiers currently disagree, and we report that rather than resolve it prematurely: on the deep panel the stopped arm degrades CRPS by about 5% while the composed arm improves it by 7 to 8%, and on the breadth tier both positions are within about 1% of the base. The deep tier at full breadth decides whether the sign flip is a long-context phenomenon or a small-panel artifact, and until then it appears in no summary of this paper.

4.4 The certificate is blind to the loss

Coverage is the certificate’s own axis, so we measure it directly. For each forecast, the 80% interval is the predictive’s 0.1 and 0.9 quantiles; empirical coverage is the share of outcomes falling inside, target 0.800. Volatility regimes classify each forecast origin by the ratio of recent to older realized volatility in its own context, so the classification uses no future information. Deep tier:

arm	overall	falling vol	flat	rising vol	spread
unconstrained	0.782	0.814	0.741	0.823	0.082
stop	0.819	0.863	0.782	0.789	0.080
compose-after	0.810	0.867	0.785	0.801	0.082

We expected the stopped arm to show good overall coverage and poor regime-conditional coverage. It shows neither defect: its coverage is the best in the table, overall and within every regime, with the same regime spread as the other arms. The arm losing the most log score is the best calibrated. No claim of miscalibration can be made against the conformal convention here, and none is. The certificate tracks the rank of the test score among pooled scores; the floor tracks conditional information the frozen score discarded; the two move independently, and a practice that monitors only coverage will never see the loss.

5 Discussion

The title is a description. The conformal recipe stops at its empirical map the way training stops at an epoch: except that in learning the stop is chosen against validation evidence, and here it is chosen against nothing. Measured, the stop costs 0.15 to 0.18 nats net of estimation on fifteen thousand series, more when the pooled window discards more, and composing an ordinary model past the map recovers a large fraction at no cost to empirical coverage. The floor identity locates the part of the loss that is fixed before calibration begins, and Remark 2 states the uncomfortable corollary: the method is deployed exactly when its floor is positive.

The recommendation is not a better calibrator, since every calibration choice moves only the estimation term. It is to treat the empirical map as a coordinate change with a certificate attached at its position, and to keep modelling on its output. The composed arm is one constructive instance, and in the sense of Remark 2 it is conformal prediction with the modelling finished; its scored calibration stream then measures, rather than assumes, whether the modelling ever was.

Limitations. The count is a keyword screen with full-text verification of every label that could change the conclusion; the 599 stop default rests on the screen plus a stated refutation criterion. The prices are measured on one base forecaster and one economic corpus plus a synthetic control; the weekly arm and the enlarged deep tier are still filling, and every constant above carries a store tag so the final version is a substitution, not a reanalysis. The composed arm has no coverage guarantee, which is the thing being priced, and it is not proposed as a conformal method. The floor proposition covers invertible signed scores only; folded and quantile scores need their own treatment.

A Proof of Proposition 1

Fix the fold, so (A, κ) is deterministic. Given $X = x$, change of variables gives the true density $p(y|x) = p_W(A(x, y)|x) A'(x, y)$, and the conformal density is $\hat{f}_x(y) = \hat{g}_{\kappa(x)}(A(x, y)) A'(x, y)$ with $\hat{g}_{\kappa}$ the density of the absolutely continuous estimate $\hat{G}_{\kappa}$. The Jacobian cancels in the log ratio, so

$$\text{KL}(P(\cdot|x) \parallel \hat{F}_x) = \text{KL}(P_W(\cdot|x) \parallel \hat{G}_{\kappa(x)}) = \text{KL}(P_W(\cdot|x) \parallel G_{\kappa(x)}) + \mathbb{E}_{W \sim P_W(\cdot|x)} \log \frac{g_{\kappa(x)}(W)}{\hat{g}_{\kappa(x)}(W)}.$$

Averaging over X : the first term is $\mathbb{E}_X \text{KL}(P_W(\cdot|X) \parallel P_W(\cdot|\kappa(X))) = \text{I}(W; X|\kappa)$ by definition, and the second averages to $\mathbb{E}_{\kappa} \text{KL}(G_{\kappa} \parallel \hat{G}_{\kappa}) \geq 0$. Every calibration operation is a function of the pooled scores alone, hence acts only on $\hat{G}_{\kappa}$ and the second term. $\square$

B Reproduction

Every constant regenerates from a tidy store keyed (`task`, `source`, `dataset`, `uid`, `model`, `horizon`, `repr`, `metric`, `value`). Breadth tier rows are tagged `forecast:CONFB:ctx256:.../m6+hash`, deep tier `CONF:ctx1024`, controls `CONFNB/CONFNULL`; the hash covers the harness sources, the tracked diff, and the compiled core. The survey classifier, frame, and per-paper verdicts live beside the grammar paper. Study declarations and the fill protocol are in the `skaters` repository (`surrogate/bench/studies.py`, `skills/run-a-study.md`).

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